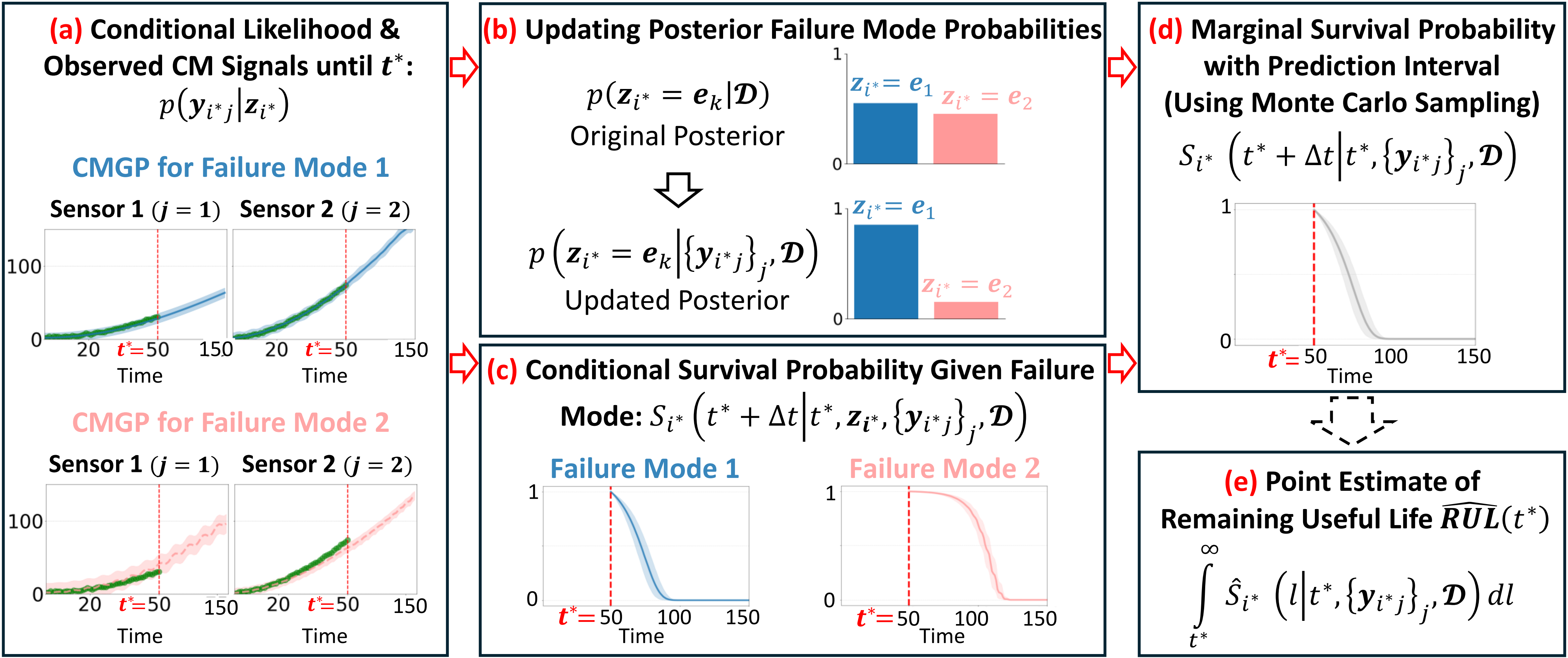

(a) Conditional Likelihood & Observed CM Signals until t^*: $p(\boldsymbol{y}_{i^*j}|\boldsymbol{z}_{i^*})$
CMGP for Failure Mode 1
Sensor 1 $(j=1)$
Sensor 2 $(j=2)$
100
0
20
$t^*=50$
150
Time
CMGP for Failure Mode 2
(b) Updating Posterior Failure Mode Probabilities
$p(\boldsymbol{z}_{i^*}=\boldsymbol{e}_k|\mathcal{D})$
Original Posterior
$p\left(\boldsymbol{z}_{i^*}=\boldsymbol{e}_k\middle|\{\boldsymbol{y}_{i^*j}\}_j,\mathcal{D}\right)$
Updated Posterior
$\boldsymbol{z}_{i^*}=\boldsymbol{e}_1$
$\boldsymbol{z}_{i^*}=\boldsymbol{e}_2$
1
0
(c) Conditional Survival Probability Given Failure Mode: $S_{i^*}\left(t^*+\Delta t\middle|t^*,\boldsymbol{z}_{i^*},\{\boldsymbol{y}_{i^*j}\}_j,\mathcal{D}\right)$
Failure Mode 1
Failure Mode 2
$t^*=50$
100
150
Time
(d) Marginal Survival Probability with Prediction Interval (Using Monte Carlo Sampling)
$S_{i^*}\left(t^*+\Delta t\middle|t^*,\{\boldsymbol{y}_{i^*j}\}_j,\mathcal{D}\right)$
(e) Point Estimate of Remaining Useful Life $\widehat{RUL}(t^*)$
$\int_{t^*}^{\infty}\hat{S}_{i^*}\left(l\middle|t^*,\{\boldsymbol{y}_{i^*j}\}_j,\mathcal{D}\right)dl$